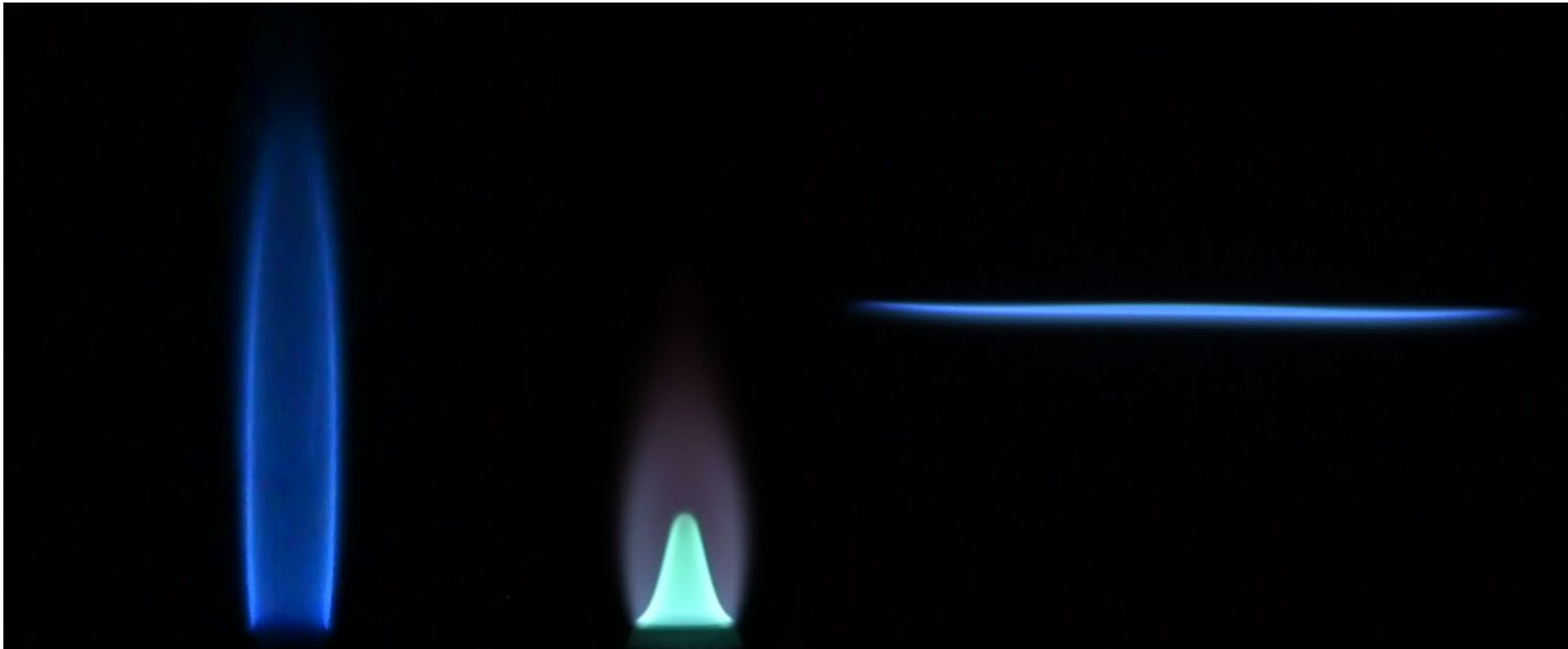


실습 가이드: OpenFOAM Laminar reacting flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Laminar reacting flow

- Laminar reacting flow



Turbulent reacting flow

• Governing equations of multiComponentFluid

- A solver for transient, compressible, **turbulent** reacting flow included in the standard OpenFOAM framework.
- Here, the **unity Schmidt number** (Sc) and **unity Lewis number** (Le) assumptions are applied for simplification.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{v} Y_i) = \nabla \cdot (\mu_{\text{eff}} \nabla Y_i) + w_i$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) = \frac{\partial p}{\partial t} + \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \sum h_i^0 w_i.$$

$$Sc = \frac{\nu}{D} = \frac{\text{kinematic viscosity}}{\text{mass diffusivity}}$$

$$Le = \frac{\alpha}{D} = \frac{\text{thermal diffusivity}}{\text{mass diffusivity}}$$

$$\mu_{\text{eff}} = \mu_l + \mu_t = \mu_l + \rho \nu_t$$

Laminar reacting flow

● Governing equations of DTLreactingFoam solver [*]

- A solver for transient, compressible, **laminar** reacting flow included in the DTLreactingFoam framework.
- Solves governing equations that account for species and energy transport due to **molecular diffusion**.
- Accurate evaluation of transport properties using a **detailed transport model (DTM)** is **essential**.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) + \nabla \cdot (\rho Y_i \mathbf{V}_i) = w_i$$

$$\begin{aligned} & \frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) - \frac{\partial p}{\partial t} \\ &= \nabla \cdot \left(\frac{\lambda}{c_p} \nabla h \right) - \nabla \cdot \left(\rho \sum_{i=1}^N h_i Y_i \mathbf{V}_i \right) + \sum h_i^0 w_i \end{aligned}$$

- Diffusion velocity

$$\mathbf{V}_i = -\frac{D_{i,\text{mix}}}{X_i} \nabla X_i$$

Laminar reacting flow

- **Detailed Transport Model (DTM) [*]**

- Transport properties such as viscosity μ , thermal conductivity λ , and binary diffusivity D_{ji} are evaluated based on **standard kinetic theory** [*]

- ❖ Viscosity

$$\mu_k = \frac{5}{16} \frac{\sqrt{\pi m_k k_B T}}{\pi \sigma_k^2 \Omega^{(2,2)*}}$$

- ❖ Thermal conductivity

$$\lambda_k = \frac{\mu_k}{m_k} (f_{\text{trans}} C_{v,\text{trans}} + f_{\text{rot}} C_{v,\text{rot}} + f_{\text{vib}} C_{v,\text{vib}})$$

- ❖ Binary diffusion coefficient

$$D_{ji} = \frac{3}{16} \frac{\left(\frac{2 \pi k_B^3 T^3}{m_{ji}} \right)^{1/2}}{p \pi \sigma_{ji}^2 \Omega^{(1,1)*}}$$

Laminar reacting flow

- **Detailed Transport Model (DTM) [*]**

- Mixture properties are obtained based on a **mixture-averaged model** [*]

- ❖ Viscosity

$$\mu = \sum_{k=1}^K \frac{X_k \mu_k}{\sum_{j=1}^K X_j \Phi_{kj}} \quad \text{where} \quad \Phi_{kj} = \frac{1}{\sqrt{8}} \left(1 + \frac{m_k}{m_j} \right)^{-\frac{1}{2}} \left(1 + \left(\frac{\mu_k}{\mu_j} \right)^{\frac{1}{2}} \left(\frac{m_k}{m_j} \right)^{\frac{1}{4}} \right)^2$$

- ❖ Thermal conductivity

$$\lambda = \frac{1}{2} \left(\sum_{k=1}^K X_k \lambda_k + \frac{1}{\sum_{k=1}^K X_k / \lambda_k} \right)$$

- ❖ Mass diffusivity

$$D_{i,\text{mix}} = \frac{1 - Y_i}{\sum_{j \neq i}^K X_j / D_{ji}} \quad \text{where} \quad D_{ji} = \frac{3}{16} \frac{\left(\frac{2 \pi k_B^3 T^3}{m_{ji}} \right)^{1/2}}{p \pi \sigma_{ji}^2 \Omega^{(1,1)*}}$$

Laminar reacting flow

● Existing framework for laminar reacting flow

- laminarSMOKE¹
- OF-Cantera:
 - reactingDNS²,
 - CTreatingFoam³,
 - DeepFlame⁴
 - combustionFoam⁵
- reactingFoam-SCI⁶,
- laminarReactingFoam⁷
- standardReactingFoam⁸

- [1] A. Cuoci, A. Frassoldati, T. Faravelli, E. Ranzi, Combust. Flame 160 (2013) 870-886.
- [2] S. Zhong, F. Zhang, Z. Peng, F. Bai, Q. Du, Fuel 234 (2018) 1044-1054.
- [3] D. Zhou, H. Zhang, S. Yang, Aerospace 9 (2022).
- [4] R. Mao, M. Lin, Y. Zhang, T. Zhang, Z.-Q. J. Xu, Z. X. Chen, Comput. Phys. Commun. 291 (2023) 108842.
- [5] T. Li, J. Pan, F. Kong, B. Xu, X. Wang, Comput. Fluids 213 (2020) 104718.
- [6] Q. Yang, P. Zhao, H. Ge, Comput. Fluids 190 (2019) 114-127.
- [7] A. Dasgupta, E. Gonzalez-Juez, D. C. Haworth, Comput. Phys. Commun. 237 (2019) 219-229.
- [8] F. Wang, L. An, T. L. Chan, Comput. Phys. Commun. 309 (2025) 109472

Laminar reacting flow

● Limitations of existing framework

Advantage

- Provide a detailed transport model (DTM)
- Integrates with OF as a third-party package

Disadvantage

- Does not fully **support all native** OF functionality
- The lack of a **user guide** and **C++ class diagrams** limits code extensibility and flexibility.

laminarSMOKE

Advantage

- Provide detailed chemistry and transport through Cantera
- Integrates Cantera-based thermochemical models with OF

Disadvantage

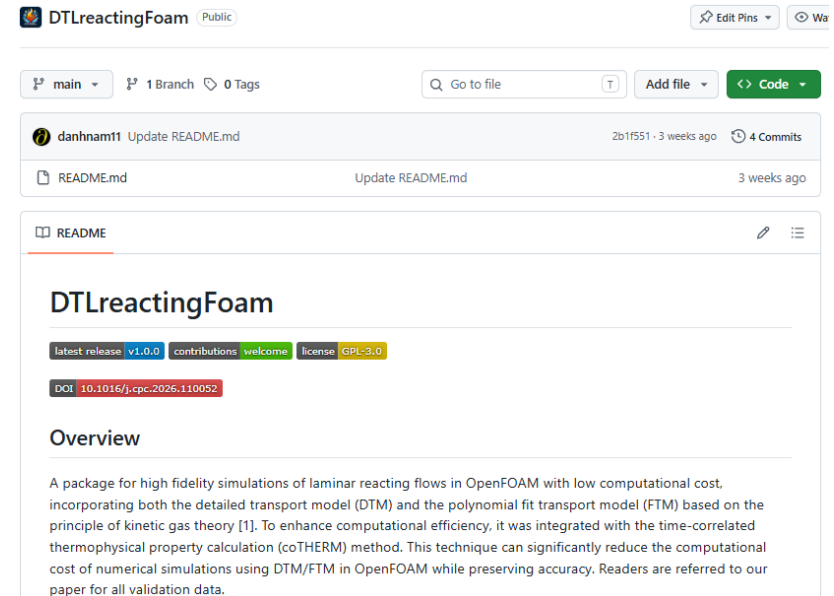
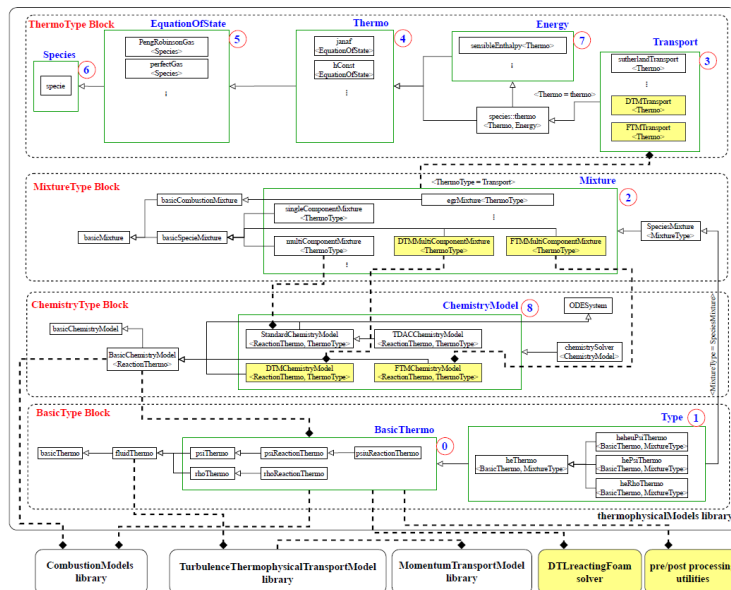
- **Dependency** on external library with **version compatibility** issues
- The **OF–Cantera interface** limits code extensibility and flexibility

Cantera coupled frameworks

Laminar reacting flow

● DTLreactingFoam [*]:

- Our group developed a **fully native** OF framework for DTM-based laminar reacting flow simulation while preserving **existing OF functionality**.
- Provides methods for reducing computational cost through **FTM** and **coTHERM** strategy.
- **Source code**, **User guide** and **tutorial cases** are available on GitHub repository [**]



▪ Class diagram of DTLreactingFoam

▪ DTLreactingFoam GitHub repository [**]

Laminar reacting flow

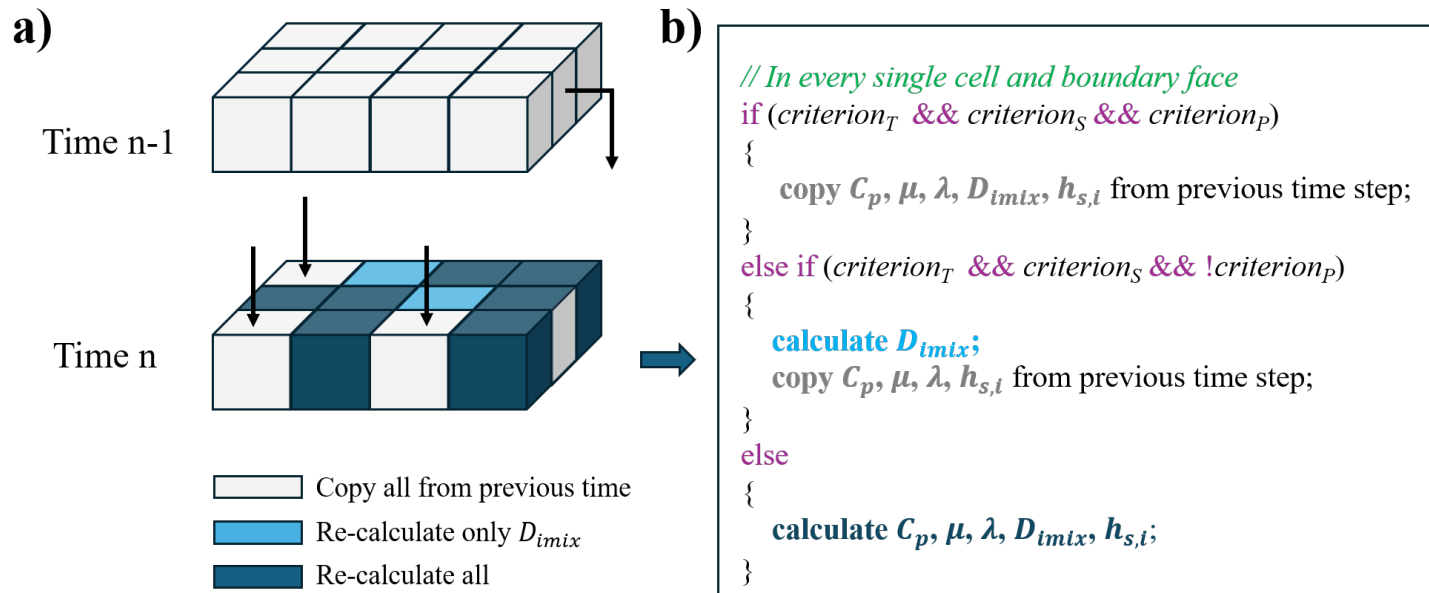
● Methods for reducing computational cost

1. Polynomial-fit transport models (FTM):

$$\ln \mu_k = \sum_{n=1}^N a_{n,k} (\ln T)^{n-1}; \quad \ln \lambda_k = \sum_{n=1}^N b_{n,k} (\ln T)^{n-1}; \quad \ln D_{jk} = \sum_{n=1}^N d_{n,jk} (\ln T)^{n-1}$$

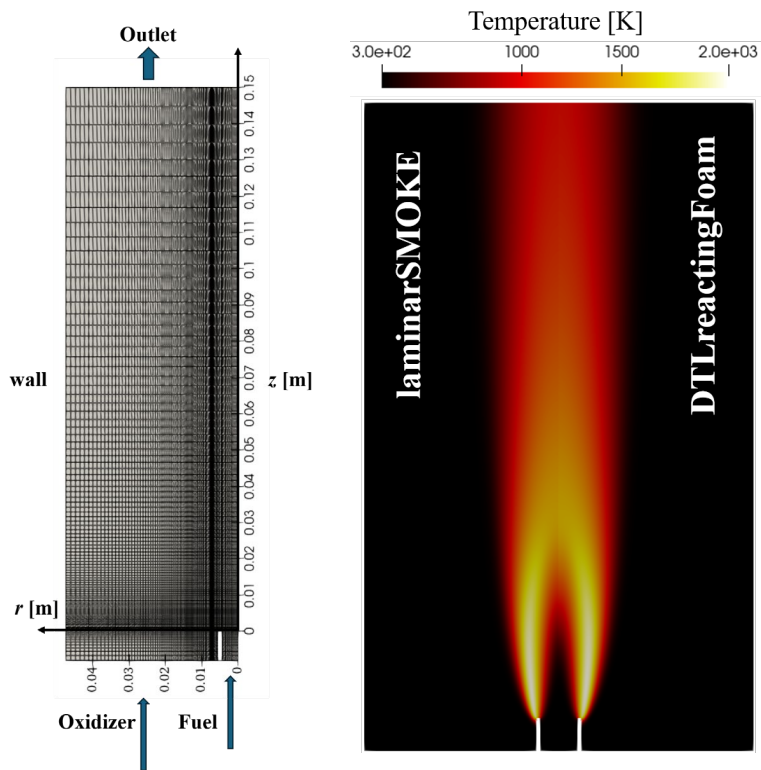
Using third-order polynomial fits: $N = 4$

2. Time correlation for thermophysical properties (**coTHERM**): [*]



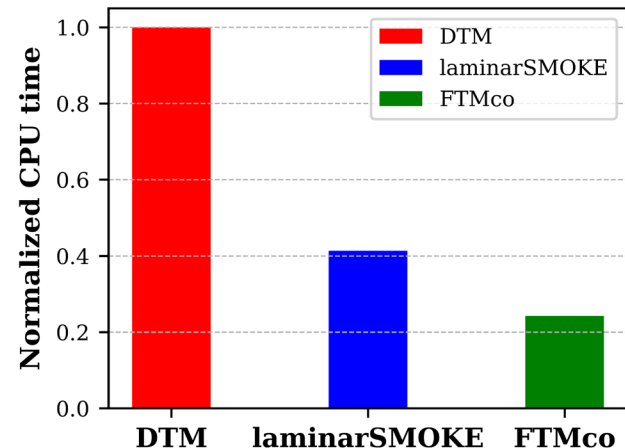
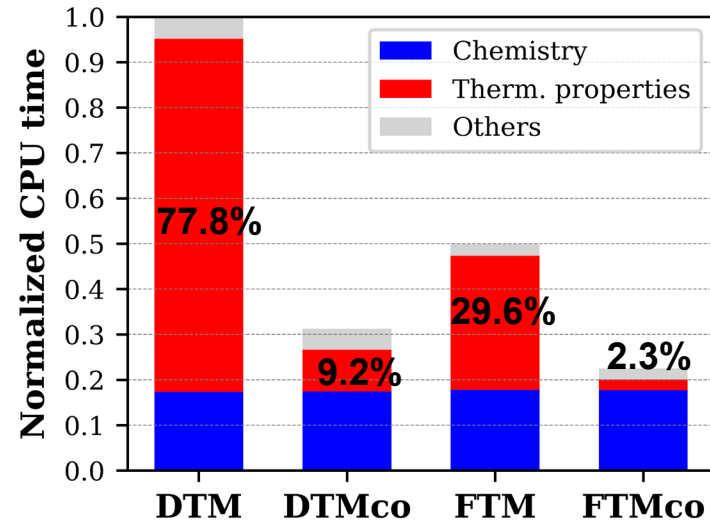
Laminar reacting flow

● Computational efficiency of DTLreactingFoam



Threshold values

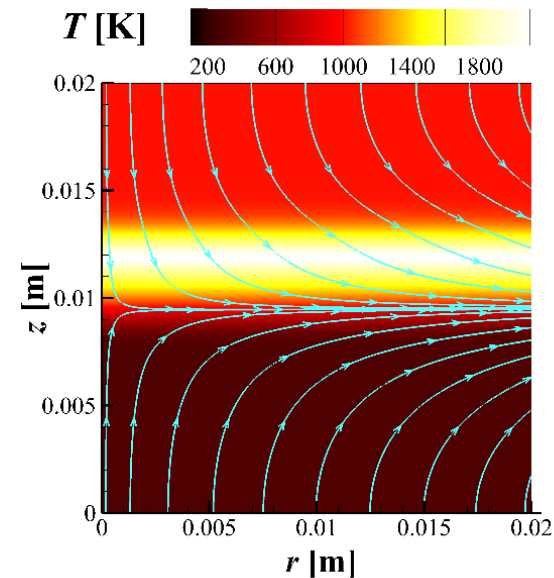
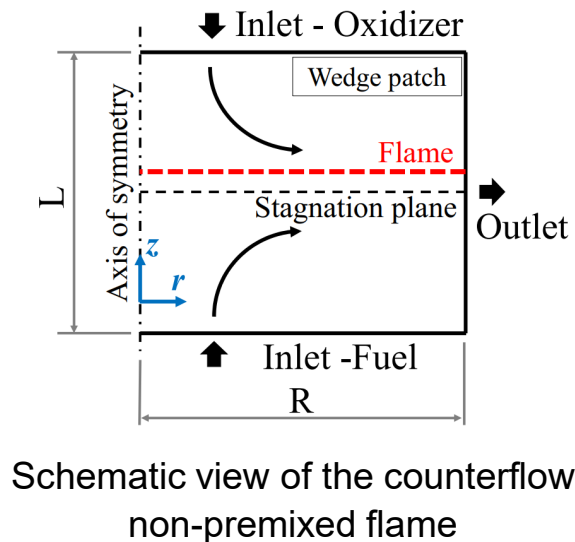
ε_T [K]	ε_S [-]	ε_p [Pa]
0.2	1E-3	100



Laminar reacting flow

● Tutorial 3: Counter flow flame

- 2D **laminar** non-premixed counter flow flame
- Fuel: 50% H_2 + 50% N_2 , Oxidizer: 21% O_2 + 79% N_2
- Inlet velocity U_{fuel} : 0.5 m/s, U_{oxi} : 0.5 m/s, Pressure: 1 atm
- Perform simulations using **DTM** and **FTM + CoTHERM (FTMco)**.



The temperature isocontours of 2-D laminar counterflow non-premixed flame

DTM model

Laminar reacting flow

- Preprocessing

- Get the tutorial files

1

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
```

```
ghkrk@20225263-p01:~$ cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ ll
total 32
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  310 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk  684 Oct  6 10:56 Allrun.DTM*
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ |
```

Laminar reacting flow

● Preprocessing: Kinetic mechanism

- Detailed hydrogen oxidation mechanism of Li et al. [*].

Table I Detailed H₂/O₂ Reaction Mechanism

	A	n	E	Reference
H₂/O₂ chain reactions				
1. H + O ₂ = O + OH	3.55×10^{15}	-0.41	16.6	[16]
2. O + H ₂ = H + OH	5.08×10^4	2.67	6.29	[44]
3. H ₂ + OH = H ₂ O + H	2.16×10^8	1.51	3.43	[45]
4. O + H ₂ O = OH + OH	2.97×10^6	2.02	13.4	[46]
H₂/O₂ dissociation/recombination reactions				
5. H ₂ + M = H + H + M ^a	4.58×10^{19}	-1.40	104.38	[47]
H ₂ + Ar = H + H + Ar	5.84×10^{18}	-1.10	104.38	[47]
H ₂ + He = H + H + He	5.84×10^{18}	-1.10	104.38	See text
6. O + O + M = O ₂ + M ^a	6.16×10^{15}	-0.50	0.00	[47]
O + O + Ar = O ₂ + Ar	1.89×10^{13}	0.00	-1.79	[47]
O + O + He = O ₂ + He	1.89×10^{13}	0.00	-1.79	See text
7. O + H + M = OH + M ^a	4.71×10^{18}	-1.0	0.00	[47]
8. H + OH + M = H ₂ O + M ^b	3.8×10^{22}	-2.00	0.00	See text
Formation and consumption of HO₂				
9. H + O ₂ + M = HO ₂ + M ^c	k_0 6.37×10^{20}	-1.72	0.52	[19]
H + O ₂ + M = HO ₂ + M ^d	k_0 9.04×10^{19}	-1.50	0.49	[19]
	k_∞ 1.48×10^{12}	0.60	0.00	[48]
10. HO ₂ + H = H ₂ + O ₂	1.66×10^{13}	0.00	0.82	[11]
11. HO ₂ + H = OH + OH	7.08×10^{13}	0.00	0.30	[11]
12. HO ₂ + O = OH + O ₂	3.25×10^{13}	0.00	0.00	[34]
13. HO ₂ + OH = H ₂ O + O ₂	2.89×10^{13}	0.00	-0.50	[34]
Formation and consumption of H₂O₂				
14. HO ₂ + HO ₂ = H ₂ O ₂ + O ₂ ^e	4.20×10^{14}	0.00	11.98	[49]
HO ₂ + HO ₂ = H ₂ O ₂ + O ₂	1.30×10^{11}	0.00	-1.63	
15. H ₂ O ₂ + M = OH + OH + M ^f	k_0 1.20×10^{17}	0.00	45.5	[50]
	k_∞ 2.95×10^{14}	0.00	48.4	[51]
16. H ₂ O ₂ + H = H ₂ O + OH	2.41×10^{13}	0.00	3.97	[47]
17. H ₂ O ₂ + H = H ₂ + HO ₂	4.82×10^{13}	0.00	7.95	[47]
18. H ₂ O ₂ + O = OH + HO ₂	9.55×10^6	2.00	3.97	[47]
19. H ₂ O ₂ + OH = H ₂ O + HO ₂ ^e	1.00×10^{12}	0.00	0.00	[52]
H ₂ O ₂ + OH = H ₂ O + HO ₂	5.8×10^{14}	0.00	9.56	

In tran.dat:
Transport parameters for DTM

Species	Geometry	epsilon/kB	sigma	Dipole	Polarizability	Zrot
! Species Geometry epsilon/kB sigma Dipole Polarizability Zrot						
! Geometry : 0 = monatomic, 1 = linear, 2 = nonlinear						
! epsilon/kB : Lennard-Jones well depth [K]						
! sigma : Lennard-Jones diameter [Angstrom]						
! Dipole : Dipole moment [Debye]						
! Polarizability : [Angstrom^3]						
! Zrot : Rotational relaxation collision number at 298 K						
AR	0	136.500	3.330	0.000	0.000	0.000
AS	0	1045.500	4.580	0.000	0.000	0.000 ! MEC
ASH	1	199.300	4.215	0.000	0.000	1.000 ! MEC
ASH2	2	229.600	4.180	0.000	0.000	1.000 ! MEC
C	0	71.400	3.298	0.000	0.000	0.000 ! *
C2	1	97.530	3.621	0.000	1.760	4.000
C2O	1	232.400	3.828	0.000	0.000	1.000 ! *
CN2	1	232.400	3.828	0.000	0.000	1.000 ! OIS
C2H	1	265.300	3.721	0.000	0.000	2.500 ! NMM
C2H2	1	265.300	3.721	0.000	0.000	2.500 ! NMM

Laminar reacting flow

- **Preprocessing: Kinetic mechanism**

- Convert CHEMKIN-format chemistry, thermodynamic, and transport data into OpenFOAM-compatible DTM thermodynamic and reaction file

②

Command Line

```
DTMchemkinToFoam \  
chemkin/chem.inp \  
chemkin/therm.dat \  
chemkin/tran.dat \  
chemkin/transportSutherland \  
constant/reactions \  
constant/thermo.DTM
```

Input

Output

Laminar reacting flow

● Preprocessing: Kinetic mechanism

- Convert Chemkin-format files into OF-format reaction and and thermodynamic file

3

Command Line

```
vi constant/thermo.DTM
```

To exit vi editor, type **:q**

:wq	write and quit the file
:w	write
:q	quit the file
:q!	quit the file (ignore the change)

```
species          9 ( H2 O2 O OH H2O H HO2 H2O2 N2 );

OH
{
  specie
  {
    molWeight      17.00737;
  }
  thermodynamics
  {
    Tlow           200;
    Thigh          6000;
    Tcommon        1000;
    highCpCoeffs   ( 2.86472886 0.00105650448 -2.59082758e-07 3.05218674e-11 -1.33195876e-15 3683.62875 5.70164073 );
    lowCpCoeffs    ( 4.12530561 -0.00322544939 6.52764691e-06 -5.79853643e-09 2.06237379e-12 3346.30913 -0.69043296 );
  }
}

transport
{
  As              1.512e-06;
  Ts              120;
  linearity        1;
  epsilonOverKb    80;
  sigma           2.75;
  dipoleMoment     0;
  alpha           0;
  Zrot             0;
}

elements
{
  O              1;
  H              1;
}
}
```

Transport parameters for DTM

Laminar reacting flow

● Preprocessing: Physical models

1. Enable detailed transport and configure CoTHERM algorithm.
2. Set the CoTHERM threshold for T , p , and Y_i .
3. Select the multicomponent mixture and detailed transport models.
4. Include thermo.DTM.

4

Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT 0.2; // [K]
epsilonP 100; // [Pa]
epsilonS 0.001; // [-], in mass fraction

thermoType
{
    type hePsiThermo;
    mixture DTMMulticomponentMixture;
    transport DTMTransport;
    thermo janaf;
    energy sensibleEnthalpy;
    equationOfState perfectGas;
    specie specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// *****
```

1

2

3

4

Laminar reacting flow

- Preprocessing: Mesh Generation

5

Command Line

blockMesh

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ blockMesh
/*-----*/
=====
\\      /  F ield      | OpenFOAM: The Open Source CFD Toolbox
\\      /  O peration  | Website:  https://openfoam.org
\\      /  A nd        | Version:  12
\\      /  M anipulation|
=====
/*-----*/
Build   : 12-86e126a7bc4d
Exec    : blockMesh
Date    : Oct 06 2026
Time    : 11:01:29
Host    : "20225263-p01"
PID     : 73088
I/O     : uncollated
Case    : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// *****
Create time
```

Laminar reacting flow

- Preprocessing: Domain decomposition
 - Parallel computing

In system/decomposeParDict file

```
/*-----* C++ *-----*/
//
// \ \ \ \ \
//   F ield      | OpenFOAM: The Open Source CFD Toolbox
//   O peration  | Website: https://openfoam.org
//   A nd        | Version: 12
//   M anipulation|
//
/*-----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// ***** //

numberOfSubdomains 4;

method          simple;

simpleCoeffs
{
    n             (1 4 1);
    delta         0.001;
}
// ***** //
```

6

Command Line

```
vi system/decomposeParDict
```

Laminar reacting flow

- Preprocessing: Domain decomposition
 - Parallel computing

7

Command Line

```
decomposePar
```

```
11
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ 11
total 52
drwxr-xr-x 10 ghkrk ghkrk 4096 Oct  7 16:56 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  6 14:18 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:56 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  684 Oct  6 10:56 Allrun*
-rw-r--r--  1 ghkrk ghkrk 2156 Oct  6 14:54 README.md
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  7 16:56 constant/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor0/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor1/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor2/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 11:03 system/
```

Domain is decomposed

Laminar reacting flow

- Preprocessing: Simulation control

8

Command Line

```
vi system/controlDict
```

```
/*----- C++ -----*/
//
//   F i e l d
//   O p e r a t i o n
//   A n d
//   M a n i p u l a t i o n
//
//   OpenFOAM: The Open Source CFD Toolbox
//   Website:  https://openfoam.org
//   Version:  12
//
/*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}
// *****

application    foamRun;
solver          DTLreactingFoam;
startFrom      startTime;
startTime      0;
stopAt         endTime;
endTime        0.1;
deltaT         1e-6;
writeControl    adjustableRunTime;
writeInterval   0.005;
purgeWrite     0;
writeFormat     ascii;
writePrecision  6;
writeCompression off;
timeFormat      general;
timePrecision   6;
runtimeModifiable true;
adjustTimeStep  yes;
```

Laminar reacting flow

- Solving: Run the simulation
 - Use DTLreactingFoam solver

⑨

Command Line

```
mpirun -np 4 foamRun -parallel
```

Laminar reacting flow

● Post-Processing

10

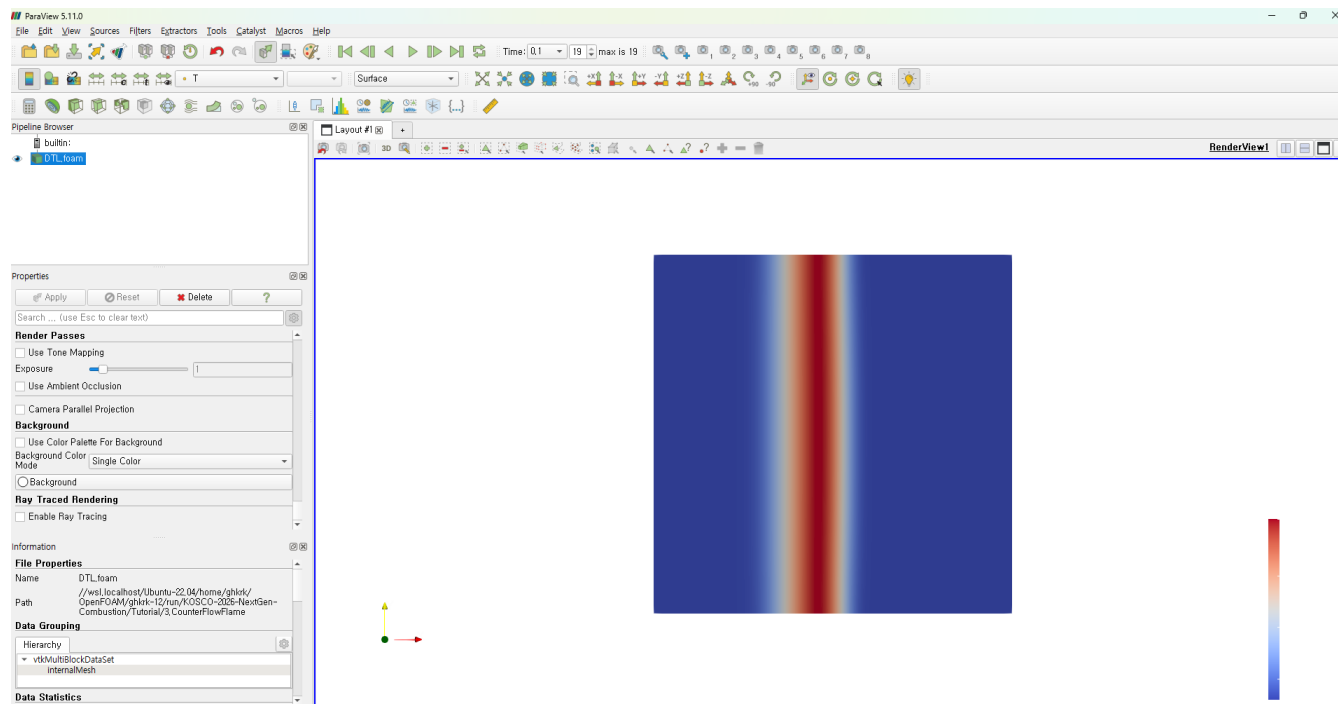
Command Line

```
reconstructPar
```

```
touch DTM.foam
```

Copy & paste files
to C: or D: drive

Open foam file
with paraview



FTM model with coTHERM (FTMco)

Laminar reacting flow

- Preprocessing for FTM

- Get the tutorial files

1

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTMC0
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTMC0$ ll
total 36
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 14:55 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 14:18 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  328 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 2451 Oct  6 14:53 Allrun*
-rw-r--r-- 1 ghkrk ghkrk 1146 Oct  6 14:55 README.md
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 14:54 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
```

Laminar reacting flow

● Preprocessing for FTM: Polynomial coefficient generation

- Enable FTM preprocessing.
- Set the transport model to preprocessingFTMTransport.

```
/*-----*-- C++ --*-----*/
//
// \ \ \ \ \ F i e l d
// \ \ \ \ \ O p e r a t i o n
// \ \ \ \ \ A n d
// \ \ \ \ \ M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website: https://openfoam.org
// Version: 12
//
/*-----*-- C++ --*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      physicalProperties;
}

// *****

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM true;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT      0.2; // [K]
epsilonP      100; // [Pa]
epsilonS      0.001; // [-], in mass fraction

thermoType
{
    type        hePsiThermo;
    mixture     DTMMulticomponentMixture;
    transport    preprocessingFTMTransport;
    thermo      janaf;
    energy       sensibleEnthalpy;
    equationOfState perfectGas;
    specie       specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// *****
```

2

Command Line

```
vi constant/physicalProperties
```

Laminar reacting flow

- Preprocessing for FTM: Polynomial coefficient generation
 - Generate thermo.FTM containing the FTM transport coefficients.

3

Command Line

```
blockMesh
```

```
FTMchemkinToFoam
```

```
vi constant/thermo.FTM
```

```
OH
{
  transport
  {
    muCoeffs      (-17.3099 1.90275 -0.170195 0.00767127);
    kappaCoeffs   (4.12383 -3.89714 0.629982 -0.0280239);
    DijCoeffs     (
      (-20.112 2.28781 -0.0847962 0.00377877)
      (-23.4526 3.14265 -0.197273 0.00872065)
      (-22.376 2.88806 -0.164634 0.00732455)
      (-9.21034 -6.44736e-26 9.25847e-27 -4.40439e-28)
      (-28.6142 5.15624 -0.446304 0.0190235)
      (-22.5274 3.44613 -0.235979 0.010369)
      (-23.4579 3.14265 -0.197273 0.00872065)
      (-23.463 3.14265 -0.197273 0.00872065)
      (-23.2325 3.04984 -0.185302 0.00820502)
    );
  }
}
```

Laminar reacting flow

● Preprocessing for FTM

- Disable FTM preprocessing and enable CoTHERM.
- Select FTMMulticomponentMixture and FTMTransport as the mixture and transport models.
- Include thermo.FTM containing the generated transport coefficients.

4

Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM true;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT      0.2; // [K]
epsilonP      100; // [Pa]
epsilonS      0.001; // [-], in mass fraction

thermoType
{
    type        hePsiThermo;
    mixture      FTMMulticomponentMixture;
    transport     FTMTransport;
    thermo       janaf;
    energy        sensibleEnthalpy;
    equationOfState perfectGas;
    specie        specie;
}

defaultSpecie N2;

#include "thermo.FTM"

// *****
```

Laminar reacting flow

- Solving: Run the simulation with FTM
 - Use DTLreactingFoam solver

5

Command Line

```
decomposePar
```

```
mpirun -np 4 foamRun -parallel
```

Laminar reacting flow

● Post-Processing

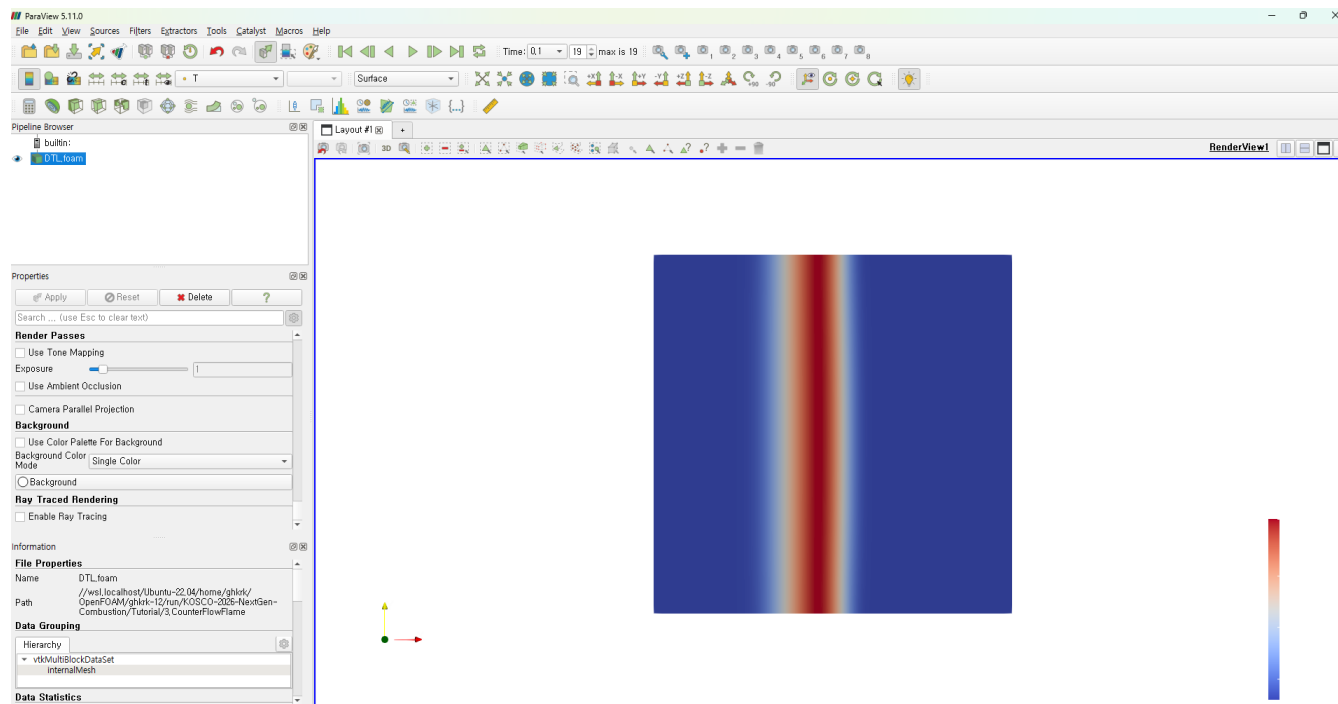
⑩ Command Line

```
reconstructPar
```

```
touch FTMC0.foam
```

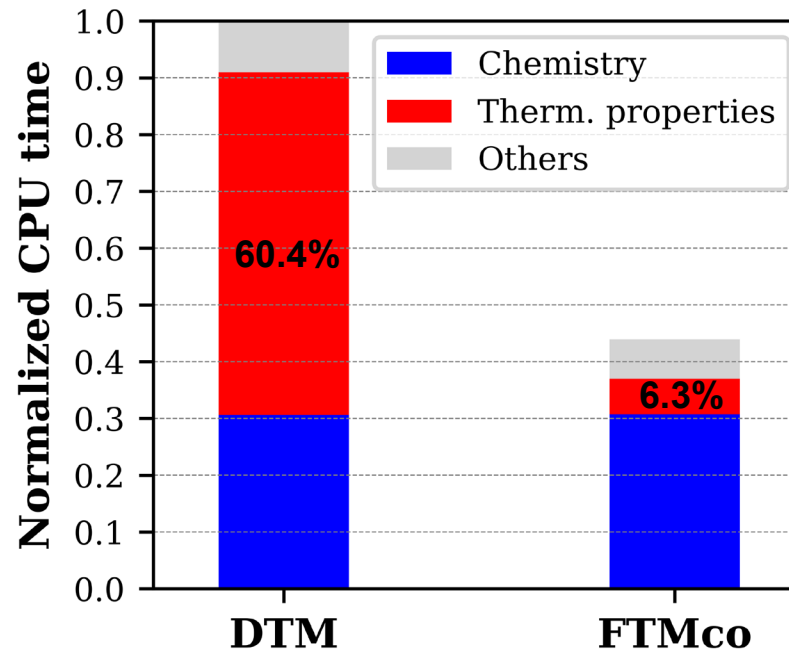
→ Copy & paste files
to C: or D: drive

→ Open foam file
with paraview



Laminar reacting flow

- Computational efficiency for CounterFlowFlame Case



CPU time: 56%
reduction